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Special Collection:

Unraveling Multi-Scale Tropical Cyclone Variabilities: Linking Mechanism, Modeling, Prediction and Projection

Key Points:

- Mixed Rossby-gravity (MRG) disturbances that transformed into TD-type disturbances are distinguished using a theoretical identification and clustering method
- Transformed MRG waves exhibit reduced vertical tilt, deeper poleward convection, and stronger moisture anomalies than non-transformed ones
- Northward moisture advection promotes off-equatorial convection, favoring the MRG waves' northward deviation and MRG-TD transition

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Moisture Characteristics Distinguishing Transformed From Non-Transformed Mixed Rossby-Gravity Waves Over the Western North Pacific

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Abstract Mixed Rossby-gravity (MRG) waves are essential synoptic-scale variability in the equatorial atmosphere. This study investigates the role of moisture variation in the transition of MRG waves to tropical depression (TD)-type waves in the western North Pacific. Using ERA5 reanalysis data, an identification algorithm based on the theoretical solution and the K-means clustering method, MRG wave events were separated into transformed (MRG-TD) and non-transformed (MRG) types. Before transforming into a TD-type disturbance, the MRG waves in the MRG-TD type exhibit stronger moisture anomalies and embedded convection compared to the MRG type, which favors intensification and the genesis of TD-type disturbance. Enhanced convection and moisture anomalies caused by a more pronounced anomalous meridional wind are located relatively northward in MRG-TD type. These northward enhanced humidification facilitates the poleward shifting of the convection and coupled circulation. The background environment also contributes to the observed differences in moisture budgets between these types, with enhanced background moisture exhibited in the lower-to-mid troposphere with stronger background ascending motion in the equatorial North Pacific. These findings provide insights into the coupling between moisture, convection, and dynamics of the MRG-TD transition.

Plain Language Summary Mixed Rossby-gravity (MRG) waves are important weather systems along the equator that move northward and transform into another system called tropical depression (TD)-type waves under favorable conditions, which usually result in typhoon genesis in the western North Pacific. In order to understand the role of moisture in this transition, we used data analysis techniques to separate MRG waves into two types: those that transformed and those that did not transform, and then we compared their differences. We found that the transformed MRG waves had a more humid and stronger upward motion environment, with enhanced northward wind inside the MRG waves, leading to more moisture and rainfall in the north of the MRG waves. Our findings demonstrate that moisture is a key factor in the evolution of MRG waves into TD-type waves, and can improve the predictive capabilities of tropical weather and the early warning of typhoon genesis.

1. Introduction

Mixed Rossby-gravity (MRG) waves are one of the fundamental equatorial wave modes in both the atmosphere and the ocean. They are westward-propagating, synoptic-scale disturbances that were first theoretically described with the linearized shallow-water equations on an equatorial beta plane by Matsuno (1966) and were subsequently observed in the stratosphere by Yanai and Maruyama (1966). MRG waves are commonly observed over the central Pacific, influencing the circulation and precipitation over the western Pacific (S.-H. Huang et al., 2022). As these waves propagate northwestward away from the equator, their wavelength occasionally shrinks from about 8,000 km to 4,000 km, and they eventually transform into a series of cyclonic and anticyclonic vortices with enhanced or suppressed convection, commonly referred to as tropical depression (TD)-type disturbances (Dickinson & Molinari, 2002; Nitta, 1970; Takayabu & Nitta, 1993; Wallace, 1971). During the transition from MRG waves to TD-type waves, the equatorial clockwise or counterclockwise circulation associated with the MRG wave deviates northward and becomes a cyclonic or anticyclonic center. The convection center, located at the convergence zone of the MRG wave's cross-equatorial meridional wind, shifts forward and northward, overlaps with the circulation center and thereby forms a TD-type wave. TD-type disturbances serve as favorable initial disturbances for tropical cyclone (TC) formation over the western Pacific warm pool, and nearly 50% of TC genesis is associated with MRG or TD-type waves (Chen & Chou, 2014; Frank & Roundy, 2006; Wu &

Takahashi, 2018). Early TC prediction can be facilitated by this relationship through the observation of TD-type waves or their precursors—MRG waves. Therefore, understanding the factors influencing the transition from MRG waves to TD-type waves is essential for improving predictions of TC genesis.

Previous studies collectively highlight the importance of the monsoon trough's dynamical environment, including shear, convergence, and its sea surface temperature (SST)-driven variability in shaping the evolution of MRG waves toward TD-type systems. Specifically, the horizontal shear of the background zonal wind has been identified as a primary driver of the MRG-TD transition; cyclonic shear and zonal convergence associated with the northwest Pacific monsoon trough tend to shorten the wavelength of MRG waves, ultimately leading to their evolution into TD-type disturbances (Chen & Huang, 2009; Liebmann & Hendon, 1990; Webster & Chang, 1988). Wu et al. (2015) pointed out that the activity of MRG and TD-type waves is associated with monsoon trough intensity, with enhanced wave activity observed in years characterized by a strong monsoon trough. This observed pattern is attributed to interactions between the waves and low-level ambient flow. Moreover, the ocean thermal state in the western Pacific warm pool region regulates the location of the monsoon trough, and the MRG-TD wave transition zone shifts zonally in response to this movement (Huang & Chen, 2007). Hence, the El Niño-Southern Oscillation (ENSO), a primary driver of tropical variability, significantly influences the MRG-TD transition by modulating the background environment (Chen, 2012; Chen & Tam, 2012; Wu et al., 2014). Additionally, numerical simulations have shed light on the MRG-TD transition mechanism: Kuo et al. (2001) suggested that variable background flow contributes to the transition in a nonlinear barotropic model, and Ayyer and Molinari (2003) found that MRG waves grow from off-equatorial disturbances in convergent background zonal wind shear induced by persistent large-scale heating via idealized numerical experiments. However, these dynamically focused studies fail to fully explain the MRG-TD transition, as they overlook the potential role of non-dynamical factors (e.g., moisture and convection), leaving an important research gap regarding whether dynamical processes alone are sufficient to account for the transition mechanism.

Although previous studies have consistently emphasized the dominant role of dynamical factors in the MRG-TD transition, there is a certain contradiction: growing evidence indicates that the primary energy source for TD-type waves is moisture and coupled convective latent heat release (Feng et al., 2020). This contradiction highlights the necessity of exploring the role of moisture in the MRG-TD transition. Specifically, with the advancement of satellite observations, the coupling between convection and MRG waves has been revealed, and MRG waves have been classified as one of the convectively coupled equatorial waves (CCEWs) (Kiladis et al., 2009; Wheeler & Kiladis, 1999). The convergence or divergence of the lower troposphere cross-equatorial meridional wind of the MRG wave can either promote or suppress convection, and the latent heat released by convection in turn modulates the MRG waves themselves. These tropical waves interact with convection through a positive feedback mechanism that characterizes the wave-convection interaction and can alter the propagation and stability of the wave circulation (Feng et al., 2020; Lim et al., 1990; Lindzen, 1974; Reed & Recker, 1971; Takayabu & Nitta, 1993). In recent years, the moisture mode theory has emphasized the role of moisture in the propagation and structure of various equatorial perturbations, including CCEWs and the Madden-Julian Oscillation (MJO) (Adames & Maloney, 2021; Ji et al., 2025; Khouider & Majda, 2016; Takayabu, 1994; Wang & Zhang, 2015). In addition to buoyancy fluctuations and Earth's rotational effects, moisture fluctuations govern the dynamics of a range of equatorial waves (Adames et al., 2019). The MJO and intraseasonal westward-propagating equatorial Rossby waves are consistent with the moisture mode theory, in which moisture distributions control their propagation (Adames & Maloney, 2021; Mayta et al., 2022). However, it remains unclear how tropospheric moisture influences the MRG-TD transition, as canonical MRG waves are primarily governed by buoyancy and Earth's rotational effects.

Previous studies identified MRG-TD transition events based on synoptic analysis, which primarily focused on a limited number of specific cases (Dickinson & Molinari, 2002; Takayabu & Nitta, 1993). While these studies provided valuable insights into individual events, they were unable to capture the broader statistical characteristics and variability of MRG-TD transitions over longer time scales. To address this limitation, a clustering method was applied in this study to classify different types of MRG waves over a 47-year period, offering a more objective and systematic approach for identifying the factors influencing the MRG-TD transition. In this study, we investigate moisture variation during the MRG-TD transition, focusing on the contribution of moisture to the off-equatorial deviation of MRG disturbances. The structures and moisture characteristics of these two types of MRG waves are then compared and analyzed. The moisture budget reveals how moisture evolves during MRG wave evolution and provides further insight into the MRG-TD transition. The remainder of this paper is organized

as follows. Section 2 outlines the data and methods used to identify MRG waves. Section 3 presents the characteristics of MRG and MRG-TD type waves, including their temporal variations, as well as structural and moisture properties. Section 4 employs the moisture budget to investigate the differences in moisture during the MRG-TD transition. Finally, a summary and discussion are provided in Section 5.

2. Data and Methods

2.1. Data

This study employed ERA5 reanalysis data from the European Centre for Medium-Range Weather Forecasts (ECMWF), including zonal and meridional winds, vertical velocity, and specific humidity (Hersbach et al., 2023). All variables are available at a horizontal resolution of 0.5° and are sampled four times daily, covering the period from 1979 to 2025.

Precipitation rates used in this study are obtained from three data sets: the Global Precipitation Measurement (GPM) mission (Huffman et al., 2023), the Multi-Source Weighted-Ensemble Precipitation (MSWEP) product developed by Global Hydrology & Climate Center (GloH2O) (X. Wang et al., 2026), and the ERA5 reanalysis. These precipitation data serve as a proxy for convective activity over the tropical ocean and are interpolated onto a 0.5° grid. GPM data are available from June 2000 to December 2020, while MSWEP data span from 1979 to the present, and the same analysis period as the ERA5 data is used for MSWEP in this research. The results obtained from GPM in this study are generally consistent with those from MSWEP; therefore, only the results from MSWEP are presented in this paper. It should be noted that to ensure the consistency and conservation of variables, ERA5 precipitation data are employed in the moisture budget analysis performed in this study.

The 0.5° horizontal resolution employed in this study is sufficient to resolve synoptic-scale MRG and TD-type waves while also balancing computational efficiency. Meanwhile, the Niño 3.4 index is used to analyze the relationship between MRG wave events and the ENSO. The index is calculated from ERSSTv5 data provided by the NOAA Physical Sciences Laboratory. In addition, the International Best Track Archive for Climate Stewardship (IBTrACS) data set is utilized to reveal the correlation between MRG wave and TC genesis (Gahtan et al., 2024; Knapp et al., 2010).

2.2. Identification of MRG Wave Events

A method for identifying MRG wave events was developed in this study. By filtering the 850 hPa meridional wind field to retain 3–8-day westward-propagating fluctuations with zonal wavenumber k ranging from -20 to 0 , we isolated signals associated with MRG and TD-type waves. MRG waves are further identified by their characteristic cross-equatorial symmetric meridional wind structure. The identification method employed in this study is adapted from the approach proposed by Au-Yeung and Tam (2018), with some modifications to improve the identification accuracy of transformed MRG waves.

The theoretical meridional wind profile of MRG waves along the equator can be written as

$$v_{\text{fitted}}(x, y, t) = v_{\text{data}}(x, y = 0^\circ, t) e^{-[y/(2y_0)]^2}, \quad (1)$$

where y_0 is the trapping scale and $v_{\text{data}}(x, y = 0^\circ, t)$ represents the filtered meridional wind (3–8-day period and -20 – 0 zonal wavenumber) in the ERA5 reanalysis data. Then, considering the error between the theoretical meridional wind and the fitted meridional wind within the latitude range $[-Y, Y]$ (n grid points), a dimensionless error function $E(x, t)$ is defined as

$$E(x, t) = \frac{\sqrt{\frac{1}{n} \sum_{y=-Y}^{y=Y} [v_{\text{data}}(x, y, t) - v_{\text{fitted}}(x, y, t)]^2}}{\text{abs}[v_{\text{data}}(x, y = 0^\circ, t)]}. \quad (2)$$

For each time step, potential MRG wave disturbances are identified from the local maxima and minima of the equatorial meridional wind field $v_{\text{data}}(x, y = 0^\circ, t = t_0)$. At each of these local extrema, the parameter y_0 is determined by minimizing the error function E , and the instantaneous zonal wavenumber k is derived via wavelet

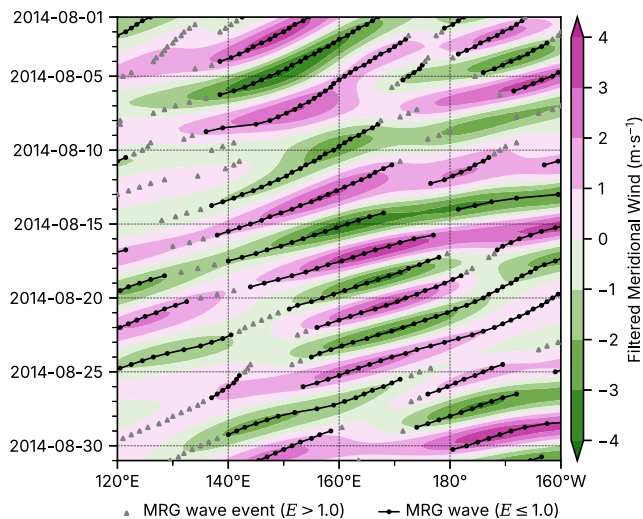


Figure 1. Hovmöller diagram illustrating the mixed Rossby-gravity (MRG) wave identification results for August 2014. The filtered meridional wind (3–8-day period and -20 – 0 zonal wavenumber) at 850 hPa, averaged over 5°S – 5°N , is shaded. Black dots and lines indicate identified valid MRG wave disturbances and individual MRG wave event tracks, respectively. Gray triangles mark invalid MRG wave disturbances that are greater than the error threshold.

individual MRG waves. To confine our analysis to the western Pacific domain and the MRG-TD transformation process, individual MRG wave events are selected according to the following two constraints. First, the center of an MRG wave must propagate continuously from east of 170°E to west of 155°E , as previous studies have demonstrated that MRG waves can undergo transformation as they propagate west of 155°E (Dickinson & Molinari, 2002; Takayabu & Nitta, 1993). Second, only MRG wave events with southerly wind phase are retained, as the convection associated with these southerly phase MRG waves will transition to the convective phase of TD-type waves if MRG-TD transformation occurs. In total, 1,287 qualified individual MRG wave events are selected over the 47-year study period for subsequent analysis.

2.3. Clustering and Composite Method

To separate distinct types of MRG waves, including non-transformed MRG waves (hereafter referred to as MRG waves or MRG type) and transformed MRG waves (hereafter referred to as MRG-TD waves or MRG-TD type), the K-means clustering algorithm is employed for this classification task (Macqueen, 1967). This method provides an objective clustering approach, free from subjective bias, by grouping samples based on their similarity.

Clustering analysis is performed based on 1,287 qualified MRG wave events selected via the identification and screening criteria introduced in the previous section. For each MRG wave event, we extract the filtered 850 hPa meridional wind field over the domain 120°E – 180° , 20°S – 20°N , sampled at 54 hr after the MRG wave crosses 160°E , as the input sample for K-means clustering. This 54 hr lag is selected based on prior studies, which have documented that the transition of MRG waves to TD-type waves completes within 2–3 days (Chen & Tam, 2012; Dickinson & Molinari, 2002; Takayabu & Nitta, 1993). Our analysis indicates that this transition process is typically completed approximately 54 hr after MRG waves pass 160°E , and this lag value maximizes the separability of transformed and non-transformed MRG wave events in the clustering. Sensitivity tests with alternative lag times (e.g., 48 hr, 72 hr) produce negligible differences in the clustering results.

The number of clusters is set to $k = 2$ for this analysis, as the 3–8-day filtering applied to reanalysis data retains primarily signals associated with MRG and TD-type waves in the study domain. Additional tests with k ranging from 3 to 7 show that larger k values only subdivide MRG or TD-type wave disturbance into subcategories with minor morphological differences, which do not alter the fundamental classification between transformed and non-transformed MRG waves but introduce unnecessary complexity to subsequent analysis. The 1287 MRG wave

analysis using the Mexican hat function. An error threshold $\hat{E}$ is defined to filter for sufficiently strong MRG wave signals: if $E \leq \hat{E}$, the corresponding local extremum is classified as a valid MRG wave disturbance.

In this study, the center of an MRG wave is defined as the location of the maximum meridional wind at the equator, corresponding to the position of the identified valid MRG wave disturbance that forms part of the MRG wave track. Unlike the method in Au-Yeung and Tam (2018), the parameter Y is set to a wider range of 20° , and the original meridional wind field, rather than its symmetric component, is used to calculate the error function. This modification increases the hit rate of the identification algorithm to MRG waves that shift off the equator, particularly for transformed MRG waves. The error threshold $\hat{E}$ determines the minimum strength above which a meridional wind disturbance is identified as an MRG wave disturbance. A range of threshold values from 0.6 to 1.2 was tested, and no significant influence on the main findings was observed. The threshold $\hat{E}$ is ultimately set to 1.0 for all analyses in this study, as this value captures a sufficient number of MRG wave disturbance samples for clustering analysis without introducing excessive noise. An example of the MRG wave event identification results is shown in Figure 1, in which only valid MRG wave disturbances are linked to form continuous tracks of individual MRG wave events.

Once all valid MRG wave events are identified, events across consecutive time steps (six-hour intervals) are linked into continuous tracks based on their temporal and spatial proximity, which represent the full life cycle of indi-

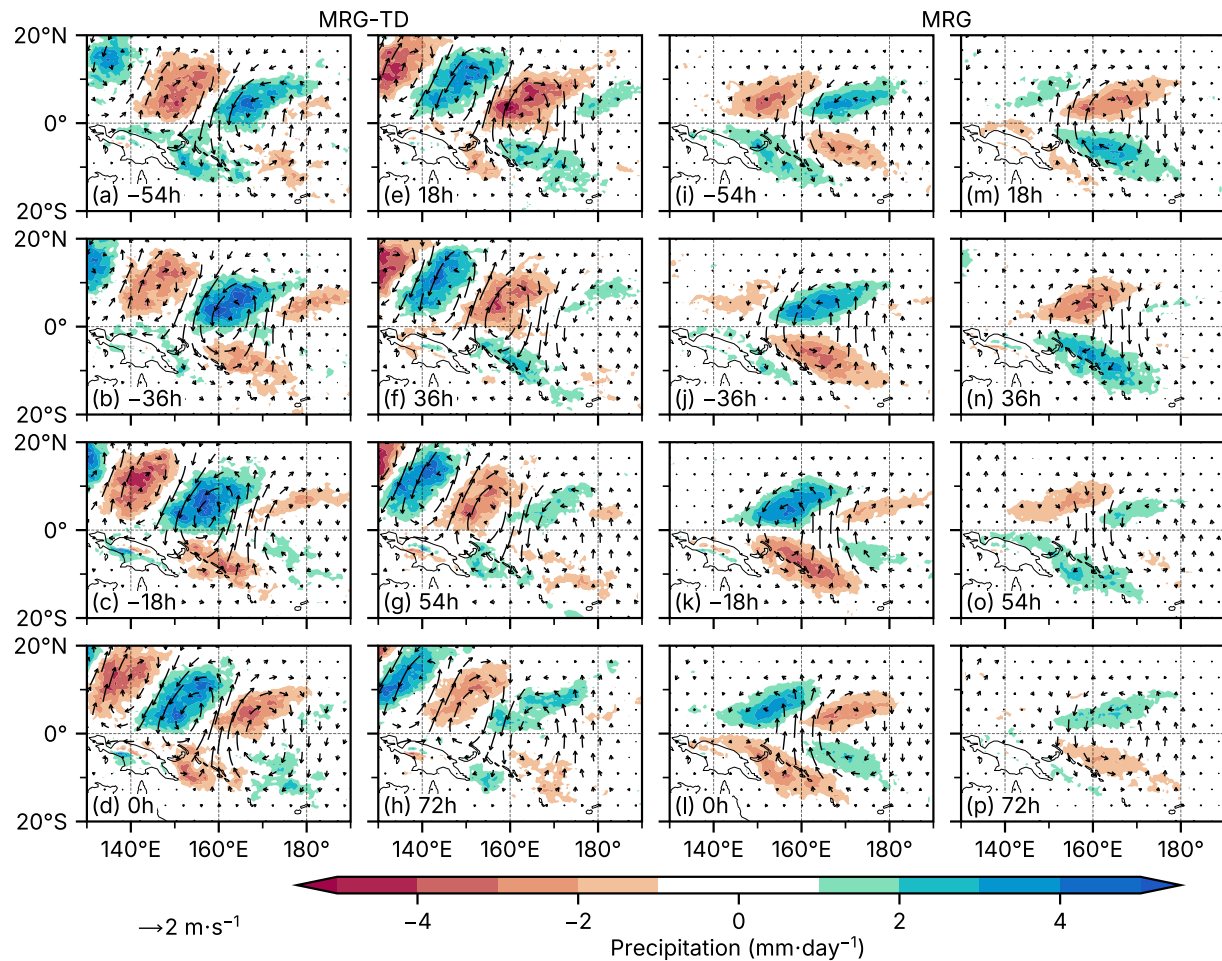


Figure 2. Lead-lag composites of 3–8-day filtered anomalous wind (vector, $\text{m} \cdot \text{s}^{-1}$) at 850 hPa and precipitation (shaded, $\text{mm} \cdot \text{day}^{-1}$) for the MRG-TD type (a–h) and mixed Rossby-gravity (MRG) type (i–p). Hours in the labels indicate lead (negative) and lag (positive) times relative to the time when the MRG wave center crosses 160°E .

samples are accordingly divided into two types, with 360 classified as the MRG-TD type and the remaining 927 classified as the MRG type.

Figure 2 presents the lead-lag composites of anomalous 850 hPa wind and precipitation for the two MRG wave types, with the 0-hr lag defined as the time when the MRG wave center crosses 160°E . At the -54 hr lag, an MRG wave disturbance is observed for both types with characteristic cross-equatorial southerly flow near 180° (Figures 2a and 2i), propagating westward gradually. After crossing 160°E , the circulation and convection associated with the MRG-TD type exhibit clear northward deviation. The zonal scale of the MRG-TD disturbance contracts during this period, while the intensity of winds and coupled convection remains stable, indicating that the MRG wave transformed into a TD-type disturbance (Figures 2d–2h). In contrast, the non-transformed MRG wave begins to weaken after crossing 160°E , and eventually dissipates after 54 hr without significant northward deviation (Figures 2l–2p).

Previous studies have shown that MRG waves over the western Pacific typically exhibit shorter wavelengths (approximately 5,000 km) compared to longer wavelengths (approximately 10,000 km) observed over the Central Pacific (Au-Yeung & Tam, 2018; Takayabu & Nitta, 1993). The wavelengths of the two types of MRG waves are different. The non-transformed MRG waves have a mean wavelength of approximately 4,500 km, while the transformed MRG-TD waves have a relatively shorter mean wavelength of approximately 3,600 km. This wavelength difference is consistent with the wavelength shrinkage during the transformation process documented in previous studies.

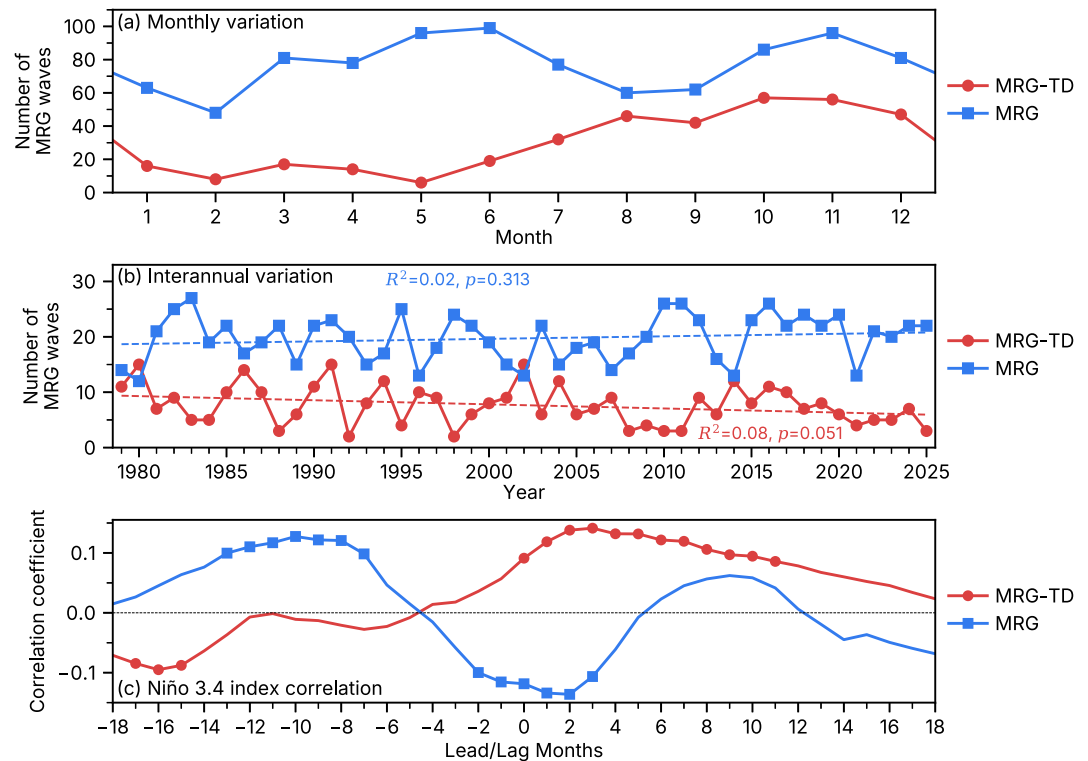


Figure 3. Temporal variations of the two types of mixed Rossby-gravity (MRG) wave during 1979–2025, including (a) monthly distribution, (b) interannual distribution, and (c) the lead-lag correlation coefficients between the monthly Niño 3.4 index and the lagged monthly number of MRG waves. The dots and squares in panel (c) indicate correlations that are statistically significant at the 95% confidence level. The statistical significance of these correlation coefficients is assessed via the Student's *t*-test.

After classification, composite analysis is performed for the two MRG wave types to characterize their structural and evolutionary differences. Since all qualified MRG waves propagate across the longitude band 155° – 170° E, we first define a set of uniformly spaced 1° longitude points spanning from 155° to 170° E. For each MRG wave event, we then linearly interpolate all relevant variables to the times when the MRG wave center reaches each of these predefined longitude points to obtain the full evolutionary process of the MRG wave event. This process not only enables us to obtain the evolution characteristics of MRG waves, but also facilitates direct comparison of the properties of different MRG waves when they arrive at the same longitude. To eliminate phase mismatches caused by variations in wavelength among individual MRG waves, we further convert the zonal longitude coordinate of each variable field to a relative zonal scale coordinate system normalized by the wavelength of the corresponding individual MRG wave event before conducting the composite. We use L to denote the half-wavelength of an MRG wave. The center of each MRG wave is set as the origin of this relative zonal scale coordinate system. Under this coordinate setup, a full wavelength of an MRG wave corresponds to a zonal scale range from $-L$ to $+L$. More detailed procedures for MRG wave tracking and compositing are described in Appendix A and Figure A1.

2.4. Moisture Budget

To obtain a complete understanding of moisture changes in the MRG-TD transition, the moisture equation is introduced, which can be written as:

$$\frac{\partial q}{\partial t} + \mathbf{V} \cdot \nabla q + \omega \cdot \frac{\partial q}{\partial p} = e - c. \quad (3)$$

In this equation, q is the specific humidity, $\mathbf{V}$ is the horizontal wind, and ω is the vertical velocity. Here, e and c denote evaporation rate and condensation rate, respectively.

Considering the background and perturbation difference, all variables are separated into two parts: the anomalous data and the background data. In equations, an arbitrary variable A can be expressed as:

$$A = A' + \bar{A}. \quad (4)$$

In this research, the background components and anomalous components are obtained by applying an 8-day low-pass filter and a 3–8-day band-pass filter to the original data, respectively. Considering the background component does not vary with time that could be omitted in the moisture equation, the anomalous moisture equation is obtained:

$$\frac{\partial q'}{\partial t} = -u' \frac{\partial \bar{q}}{\partial x} - \bar{u} \frac{\partial q'}{\partial x} - v' \frac{\partial \bar{q}}{\partial y} - \bar{v} \frac{\partial q'}{\partial y} - \omega' \frac{\partial \bar{q}}{\partial p} - \bar{\omega} \frac{\partial q'}{\partial p} + e' - c' + R. \quad (5)$$

The term R represents the residual terms that include subgrid-scale processes and errors from data. Integrating Equation 5 from 1,000 to 100 hPa, denoted by an integration symbol $\langle A \rangle = g^{-1} \int_{1000\text{hPa}}^{100\text{hPa}} A dp$:

$$\left\langle \frac{\partial q'}{\partial t} \right\rangle = -\left\langle u' \frac{\partial \bar{q}}{\partial x} \right\rangle - \left\langle \bar{u} \frac{\partial q'}{\partial x} \right\rangle - \left\langle v' \frac{\partial \bar{q}}{\partial y} \right\rangle - \left\langle \bar{v} \frac{\partial q'}{\partial y} \right\rangle - \left\langle \omega' \frac{\partial \bar{q}}{\partial p} \right\rangle - \left\langle \bar{\omega} \frac{\partial q'}{\partial p} \right\rangle + E' - P' + \langle R \rangle. \quad (6)$$

Equations 5 and 6 are the anomalous moisture budget equations for a single level and an air column, respectively. The term E' represents anomalous surface evaporation rate, P' is anomalous precipitation rate, and R is for residual terms. Within the area of convection, a major part of the upward moisture condenses into precipitation and leaves the atmosphere, hence, the vertical motion, evaporation and precipitation terms can be merged into a single term that reflects the net vertical moistening of an air column denoted by q_v :

$$q_v = -\omega' \frac{\partial \bar{q}}{\partial p} - \bar{\omega} \frac{\partial q'}{\partial p} + e' - c', \quad (7)$$

$$\langle q_v \rangle = -\left\langle \omega' \frac{\partial \bar{q}}{\partial p} \right\rangle - \left\langle \bar{\omega} \frac{\partial q'}{\partial p} \right\rangle + E' - P', \quad (8)$$

and Equation 6 can be rewritten as:

$$\left\langle \frac{\partial q'}{\partial t} \right\rangle = -\left\langle u' \frac{\partial \bar{q}}{\partial x} \right\rangle - \left\langle \bar{u} \frac{\partial q'}{\partial x} \right\rangle - \left\langle v' \frac{\partial \bar{q}}{\partial y} \right\rangle - \left\langle \bar{v} \frac{\partial q'}{\partial y} \right\rangle + \langle q_v \rangle + \langle R \rangle. \quad (9)$$

The first and third terms on the right-hand side of Equation 9 represent the horizontal advection of background moisture by anomalous winds, while the second and fourth terms represent the horizontal advection of anomalous moisture by background winds. Equations through 5–9 will be used to analyze the moisture budget of the MRG waves in the following content.

3. MRG and MRG-TD Type Characteristics

3.1. Statistical Results

Clustering results show that the number of MRG type events is nearly 2.6 times that of MRG-TD type events, indicating that the majority of MRG waves do not transition into TD-type waves. Temporal variations of the two identified MRG wave types during 1979–2025 are presented in Figure 3. MRG type waves exhibit a bimodal seasonal distribution, with peaks occurring during March–June and November–December and a minimum in February. Whereas MRG-TD type events are more active from July to December and less frequent from January to June (Figure 3a), consistent with findings from previous studies (Au-Yeung & Tam, 2018; Takayabu & Nitta, 1993). The frequency of MRG-TD type events shows a statistically significant decreasing trend over the 1979–2025 period (Figure 3b), while MRG type events show a relatively weak increasing trend. The interannual frequency variations of the two types are generally out of phase, such that an increase in one type is accompanied by a decrease in the other.

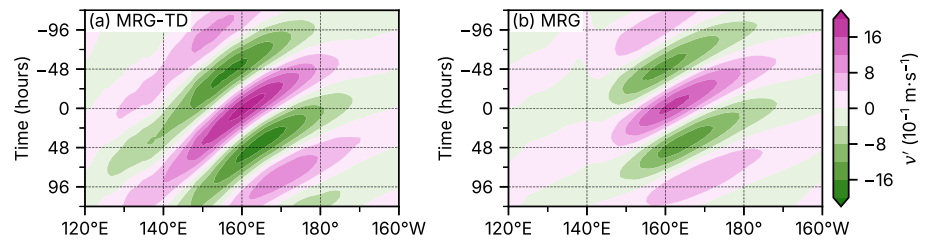


Figure 4. The Hovmöller diagram of cluster lead-lag averaged anomalous meridional wind at 850 hPa (shaded, $10^{-1} \text{ m} \cdot \text{s}^{-1}$) averaged over 10°N – 10°S for (a) MRG-TD type and (b) mixed Rossby-gravity type.

Lead-lag correlation coefficients between the monthly Niño 3.4 index and monthly events of each MRG wave type are presented in Figure 3c. MRG type events exhibit a significant negative correlation when wave activity lags the Niño 3.4 index by -2 to $+3$ months, and a significant positive correlation at lags of -13 to -7 months, indicating that MRG wave activity is not only modulated by concurrent ENSO conditions in the near term, but may also be influenced by precursory factors associated with ENSO. In contrast, MRG-TD type events show a significant positive correlation at lags of 0 to $+11$ months and a significant negative correlation at lags of -15 to -17 months, suggesting that the MRG-TD transition responds to ENSO forcing within 1 year and may also be influenced by ENSO-related precursors, similar to the MRG type. Wu et al. (2014) demonstrated that ENSO modulates the activity of both MRG and TD-type waves by altering the intensity of the monsoon trough. Enhanced (reduced) barotropic energy conversion during El Niño (La Niña) events amplifies (suppresses) TD-type disturbances, which promotes (inhibits) the transition of MRG waves to the TD-type, explaining the significant positive correlation observed for MRG-TD type waves.

3.2. Three-Dimensional Wind Structure

Figure 4 presents the propagation characteristics of the two MRG wave types in lead-lag composite Hovmöller diagrams. Prior to propagating westward to 160°E , both MRG wave types exhibit comparable westward phase speeds of approximately $-12.3 \text{ m} \cdot \text{s}^{-1}$, while the lower-tropospheric meridional wind amplitude of the MRG-TD type is notably stronger than that of the MRG type. After crossing 160°E , the disturbance for MRG type weakens gradually, whereas the disturbance for MRG-TD type sustains strong meridional wind with slower westward phase speed of approximately $-7.3 \text{ m} \cdot \text{s}^{-1}$, reflecting the transition into a TD-type wave. The group speed of the MRG-TD type is approximately $3.2 \text{ m} \cdot \text{s}^{-1}$, which is faster than that of the MRG type with the estimated $2.5 \text{ m} \cdot \text{s}^{-1}$ group speed.

Figure 5 shows the patterns of 850 hPa anomalous horizontal winds and vertical velocity for the MRG-TD type, MRG type, and their difference, composited at the same longitudes as the waves propagate westward from 170°E to 155°E (corresponding to the rightmost to leftmost columns in the figure). A distinguishing feature between the two types is that the MRG-TD type develops stronger cross-equatorial meridional winds and, north of the equator, enhanced upward motion to the west of the MRG wave center.

At 170°E (Figures 5d, 5h, and 5l), the difference in horizontal wind fields between the two types is relatively small, while the MRG-TD type exhibits significantly stronger upward motion in the region bounded by -0.5°L – 0°L and 5° – 10°N . As the MRG-TD type wave propagates further westward, the center of upward motion, which is collocated with a counterclockwise cyclonic circulation west of the MRG wave center, gradually shifts northward, with upward motion weakening while the low-level circulation intensifies (Figures 5a–5c). By the time the MRG-TD type wave propagates to 155°E , its circulation center has moved to near 8°N , deviating from the equator and completing the transition to a TD-type disturbance in subsequent days. In contrast, the counterclockwise circulation and upward motion associated with the MRG type wave remain centered near the equator and weaken gradually, eventually dispersing entirely as the wave propagates to 155°E (Figures 5e–5h). This structural difference leads to a pronounced cyclonic circulation anomaly northwest of the MRG wave center in the difference field between the two types (Figures 5i–5k), which is a classical feature of TD-type waves.

TC genesis associated with the two MRG types is visualized in Figure 5 using green dots, which mark TC genesis locations occurring within 24 hr after the MRG wave center arrives at the corresponding longitude over the 1979–2025 period (Appendix B). The ratio of TC genesis events to the total number of MRG waves differs

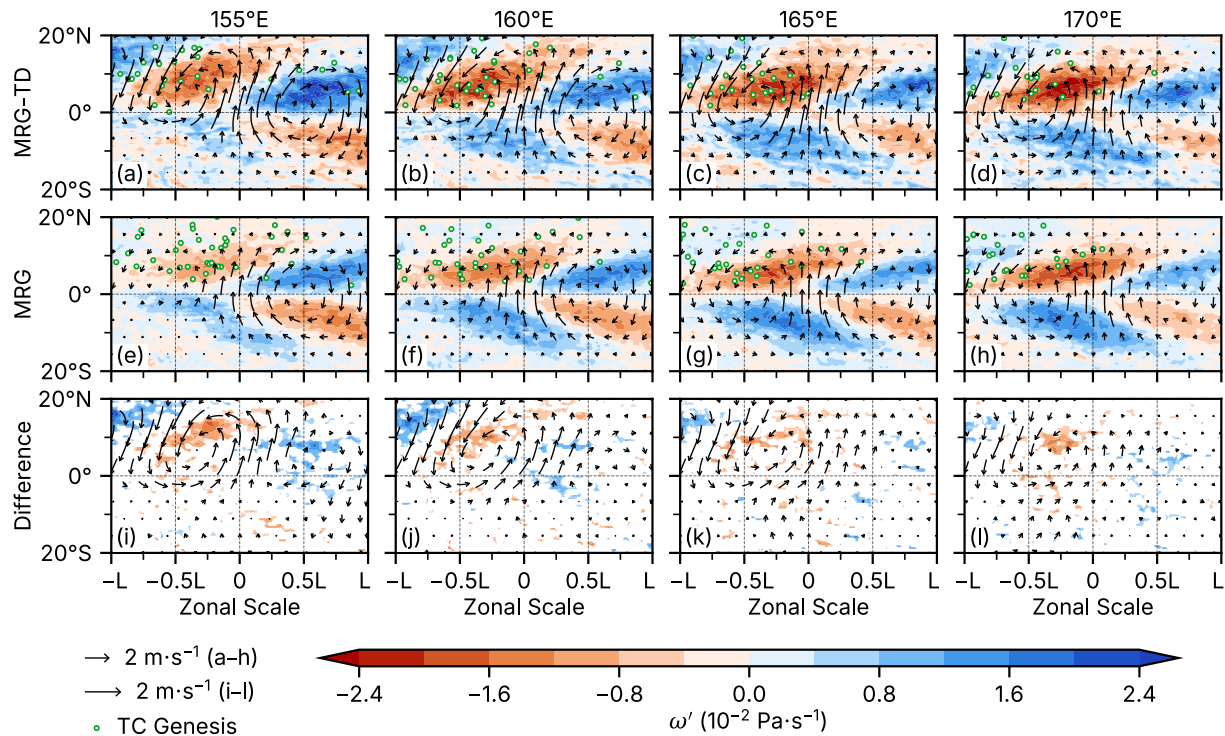


Figure 5. Anomalous horizontal winds (vector, $\text{m} \cdot \text{s}^{-1}$) and vertical velocity (shaded, $10^{-2} \text{ Pa} \cdot \text{s}^{-1}$) at 850 hPa of (a–d) MRG-TD type, (e–h) mixed Rossby-gravity (MRG) type and (i–l) their differences when MRG wave is located at (a, e, i) 155°E , (b, f, j) 160°E , (c, g, k) 165°E and (d, h, l) 170°E . Only anomalous vertical velocities that are statistically significant at the 90% confidence level are shaded in differences. The green dots indicate tropical cyclone genesis locations within 24 hr when MRG wave center is at the corresponding longitude.

significantly between the two types, as shown in Table B1. The MRG-TD type exhibits a higher TC genesis probability, with a peak ratio of 21.2% of MRG-TD waves linked to TC genesis at 160°E , whereas the MRG type reaches a maximum ratio of only 9.2% at 170°E . For the MRG-TD type, TC genesis locations are more concentrated in the upward motion center situated to the northwest of the MRG wave center. In contrast, the MRG type has a broader upward motion region, which results in a more scattered distribution of TC genesis positions. In the downward motion region located to the east of the MRG wave center, TC genesis is suppressed for both MRG types. These results are consistent with previous findings that the region to the west of the MRG wave center tends to enhance TC genesis (Frank & Roundy, 2006; Wu & Takahashi, 2018).

Figure 6 presents the vertical cross-sections of anomalous meridional wind and vertical motion for the two MRG wave types, as well as the difference fields between them. The vertical motion distributions of the two MRG wave types share similar characteristics: upward motion predominates to the west of the MRG wave center between $-0.5L$ and $0L$, spanning the entire troposphere from lower to upper levels, with its maximum amplitude occurring in the mid-to-upper troposphere. The amplified upward motion of the MRG-TD type is considerably larger than that of the MRG type, peaking in the mid-to-upper troposphere as the wave propagates to 165°E . In addition, downward motion to the east of the MRG wave center is significantly enhanced for the MRG-TD type when the wave reaches 155°E ; this enhanced downward motion is associated with the subsequent northerly phase of the MRG wave, which will later develop into the anticyclonic circulation phase of the TD-type wave. The stronger upward motion indicates that the MRG-TD type has higher convective intensity than the MRG type, which implies more latent heat release north of the equator. Notable differences are also observed in the vertical structure of meridional wind between the two types. The meridional wind amplitude of the MRG-TD type is significantly larger than that of the MRG type. Its vertical tilt reduces rapidly during westward propagation, and meridional wind extends to upper tropospheric levels. The disturbance becomes nearly vertically upright by the time the wave reaches 155°E , favoring baroclinic energy conversion (Feng et al., 2016). In contrast, eastward vertical tilting is still observed during the propagation of the MRG type.

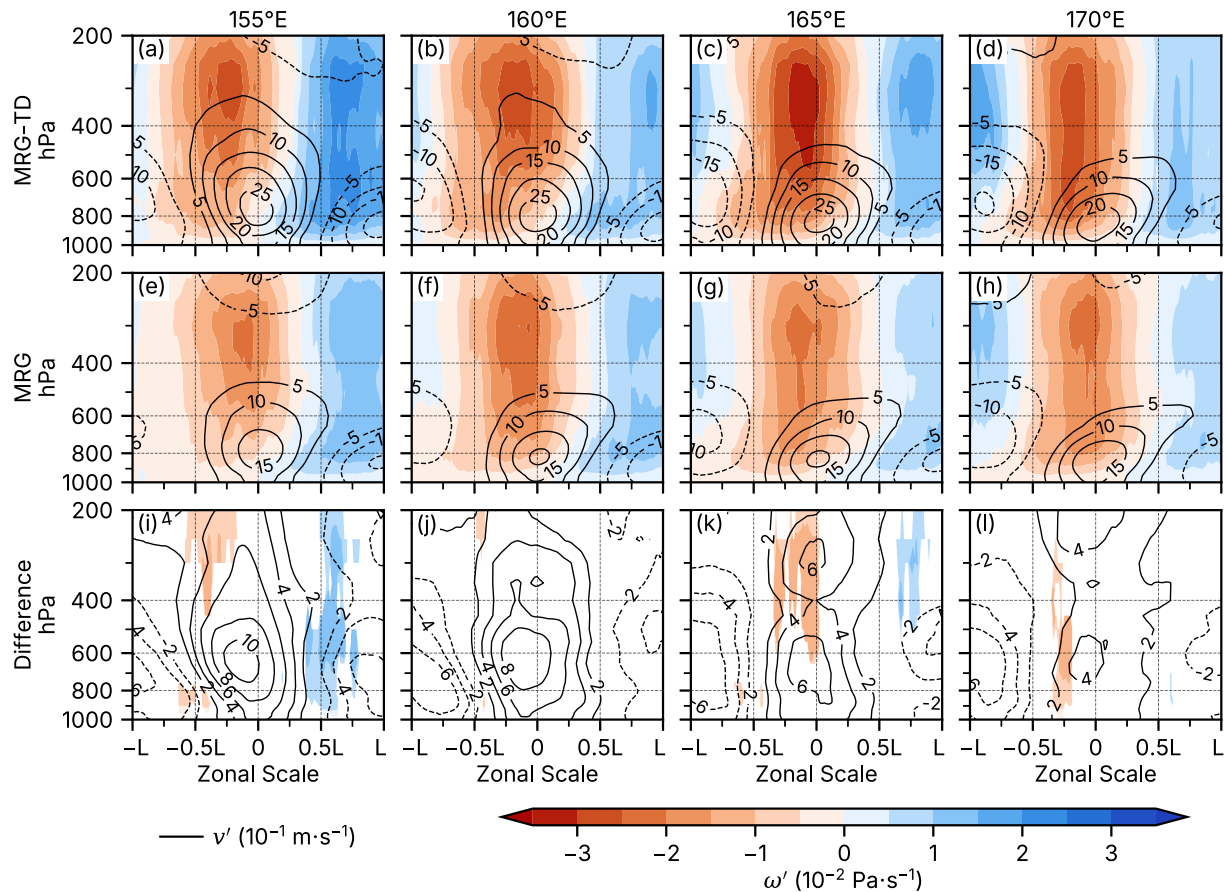


Figure 6. Anomalous vertical velocity (shaded, 10°N – 0° averaged, $10^{-2} \text{ Pa} \cdot \text{s}^{-1}$) and meridional wind (contour, 5°N – 5°S averaged, $10^{-1} \text{ m} \cdot \text{s}^{-1}$) of (a–d) MRG-TD type, (e–h) mixed Rossby-gravity (MRG) type and (i–l) their differences when MRG wave is located at (a, e, i) 155°E , (b, f, j) 160°E , (c, g, k) 165°E and (d, h, l) 170°E . Only anomalous vertical velocities that are statistically significant at the 90% confidence level are shaded in differences.

3.3. Moisture

Figure 7 presents the spatial patterns of vertically integrated anomalous moisture and anomalous moisture tendency for the two MRG wave types, alongside their difference fields. At 170°E (Figures 7d, 7h, and 7l), the anomalous moisture centers of both types are collocated near the equator (south of 5°N) to the west of the MRG wave center, with positive moisture tendency located further west of the anomalous moisture centers. MRG waves of the MRG-TD type contain substantially larger amounts of anomalous moisture, accompanied by stronger anomalous moisture tendency. Notably, the anomalous moisture tendency of the MRG-TD type extends further northward to approximately 15°N , leading to a prominent positive anomalous moisture tendency difference in the northwestern quadrant of the MRG wave center relative to the MRG type. As the MRG-TD type wave propagates westward, this northward-extended anomalous moisture tendency drives the anomalous moisture center to shift gradually northward, reaching approximately 10°N by the time the wave arrives at 155°E (Figures 7a–7d). In contrast, the anomalous moisture of the MRG type remains concentrated near the equator, with the majority of anomalous moisture confined south of 15°N , and its associated anomalous moisture tendency is weaker and restricted to the region west of the anomalous moisture center (Figures 7e–7h). These differences in the spatial distribution and magnitude of anomalous moisture and anomalous moisture tendency result in a prominent excess of anomalous moisture in the region north of the MRG wave center for the MRG-TD type relative to the MRG type, particularly after the wave propagates west of 160°E (Figures 7i–7l). This higher anomalous moisture content provides a favorable environmental condition for deep convection development and subsequent transition to TD-type waves, as TD-type waves require a certain level of moisture to sustain their convection and maintain their structure (Feng et al., 2020).

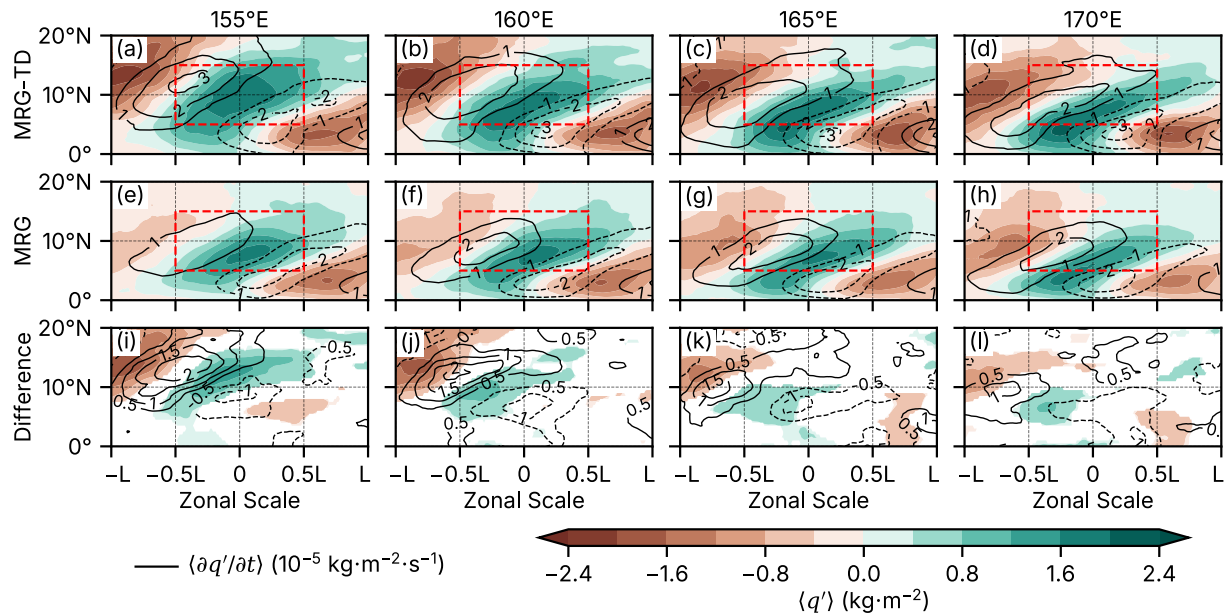


Figure 7. The spatial distribution of integrated anomalous specific humidity $\langle q' \rangle$ (shaded, $\text{kg} \cdot \text{m}^{-2}$) and integrated anomalous specific humidity tendency $\langle \partial q' / \partial t \rangle$ (contour, $10^{-5} \text{ kg} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$) of (a–d) MRG-TD type, (e–h) mixed Rossby-gravity (MRG) type and (i–l) their differences when MRG wave is located at (a, e, i) 155°E , (b, f, j) 160°E , (c, g, k) 165°E and (d, h, l) 170°E . Only anomalous specific humidity that is statistically significant at the 90% confidence level is shaded in differences. The red dashed boxes indicate the area of moisture budget calculation.

Figure 8 presents the vertical distributions of anomalous moisture and anomalous moisture tendency, averaged over 0° – 10°N , for the two MRG wave types and their corresponding difference fields. During the westward propagation of MRG waves from 170° to 155°E , the vertical tilt of anomalous moisture decreases for both types, and their vertical structures are highly similar, with the center of anomalous moisture centered in the lower-to-middle troposphere. The most prominent difference is that the MRG-TD type exhibits substantially larger anomalous moisture in the middle troposphere than the MRG type as the MRG wave propagates from 165° to 160°E , accompanied by a stronger anomalous moisture tendency in the same layer (Figures 8j and 8k). This mid-tropospheric humidification provides a more favorable environmental condition for deep convection to develop and extend to higher altitudes.

In the following section, we perform a moisture budget analysis to determine the dominant processes controlling the difference in anomalous moisture between the two types in the region north of the MRG wave center (the red dashed boxes in Figure 7), with a focus on the mechanisms driving the northward shift of anomalous moisture in the MRG-TD type.

4. Moisture Budget Results

The distribution of anomalous moisture in the MRG wave strongly influences its convective activity. By analyzing the moisture budget within the MRG wave disturbance, we can identify the factors dominating variations in anomalous moisture between the two types. The tendency of anomalous moisture is controlled by both the background conditions and the MRG wave dynamical fields, as described in Equation 5.

4.1. Background Moisture

Background conditions are one of the important factors modulating the development of MRG waves and their transition to TD-type waves. Figures 9 and 10 present the spatial distributions of 8-day low-pass filtered background flow and background moisture for both types, alongside their difference fields, when the MRG wave is centered at 160°E . The 850 hPa background wind is similar between the two types, characterized by zonal easterly flow over the tropical Pacific, with slightly weaker easterlies for the MRG-TD type, which yields a westerly anomaly in the difference field (Figures 9a–9c), particularly to the west of the MRG wave center near the equator. The distribution of background vertical velocity resembles the climatological mean precipitation pattern

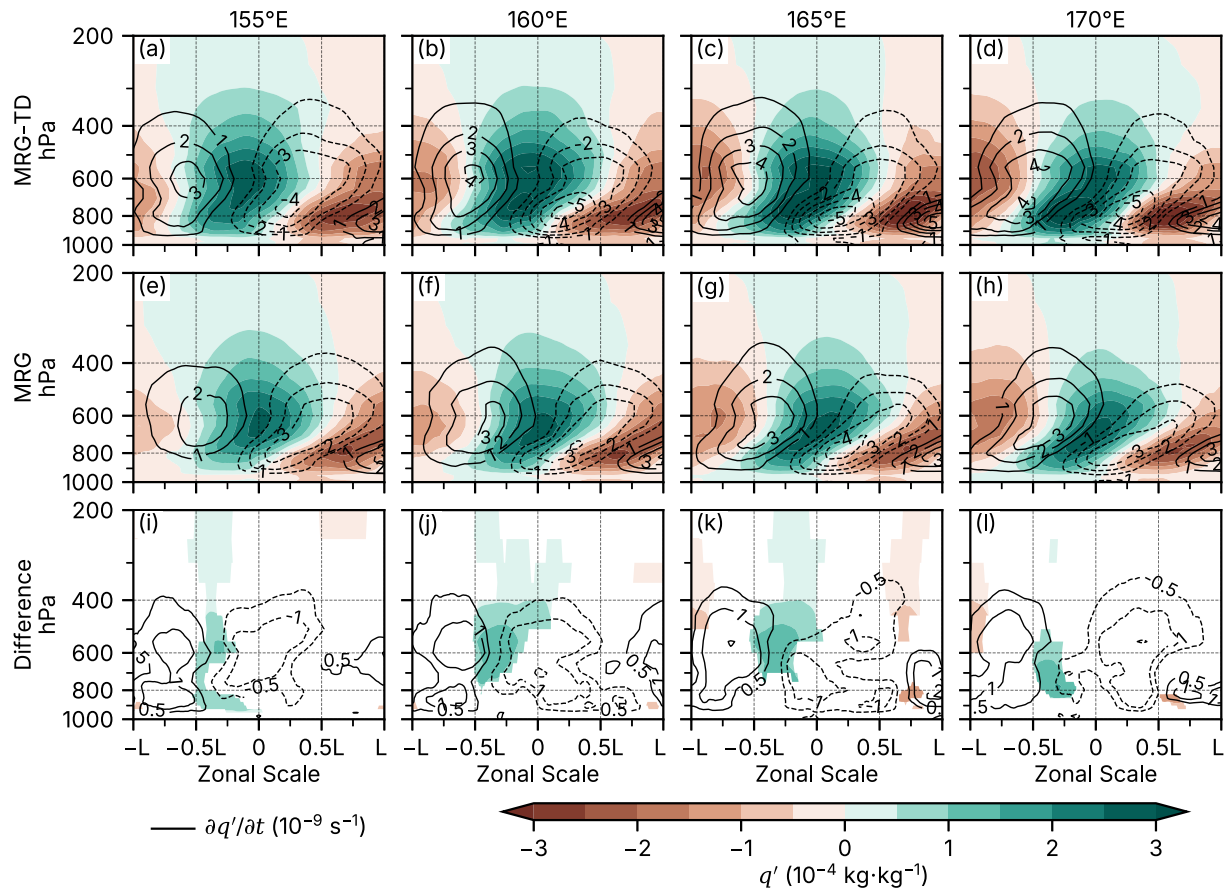


Figure 8. The 0° – 10° N averaged vertical distribution of anomalous specific humidity q' (shaded, $10^{-4} \text{ kg} \cdot \text{kg}^{-1}$) and anomalous specific humidity tendency $\partial q'/\partial t$ (contour, 10^{-9} s^{-1}) of (a–d) MRG-TD type, (e–h) mixed Rossby-gravity (MRG) type and (i–l) their differences when MRG wave is located at (a, e, i) 155°E , (b, f, j) 160°E , (c, g, k) 165°E and (d, h, l) 170°E . Only anomalous specific humidity that is statistically significant at the 90% confidence level is shaded in differences.

associated with the intertropical convergence zone, featuring a strong upward motion band between the equator and 10°N over the tropical Pacific. The background upward motion for the MRG-TD type is significantly stronger than that for the MRG type north of 10°N over the western Pacific (Figure 9c). Vertical profiles of the differences in background zonal wind and background vertical velocity over the region north of the MRG wave center are shown in Figures 10b and 10c, respectively. The difference of the background zonal wind between the MRG-TD and MRG types changes from negative to positive from 200 hPa down to 1,000 hPa, favoring an easterly shear region for the MRG-TD type. This difference in background zonal wind facilitates the reduction of vertical tilt in MRG-TD type waves. Meanwhile, significantly stronger background upward motion spanning the entire lower to upper troposphere is observed for the MRG-TD type.

The spatial pattern of background moisture features a zonal (west-east oriented) ridge near 6°N and higher moisture content over the western Pacific warm pool (Figures 9d–9f). A prominent difference is that the MRG-TD type has notably higher background moisture content north of 5°N , with a peak anomaly near 13°N , consistent with the spatial difference in background vertical velocity. This positive moisture anomaly is primarily confined to the mid-lower troposphere, with a maximum value near 700 hPa, as shown in Figure 10a. Over the region north of 5°N , the background environment for the MRG-TD type is characterized by higher humidity and enhanced upward motion in the subtropical region of the western North Pacific, which creates favorable conditions for convection development within the MRG wave.

4.2. Role of Vertical Moistening

Figure 11 illustrates the terms of the net vertical moistening defined in Equation 8 for north of the MRG wave center (5° – 15°N , normalized zonal scale $-0.5L$ to $0.5L$, as shown in the red dashed boxes in Figure 7). The

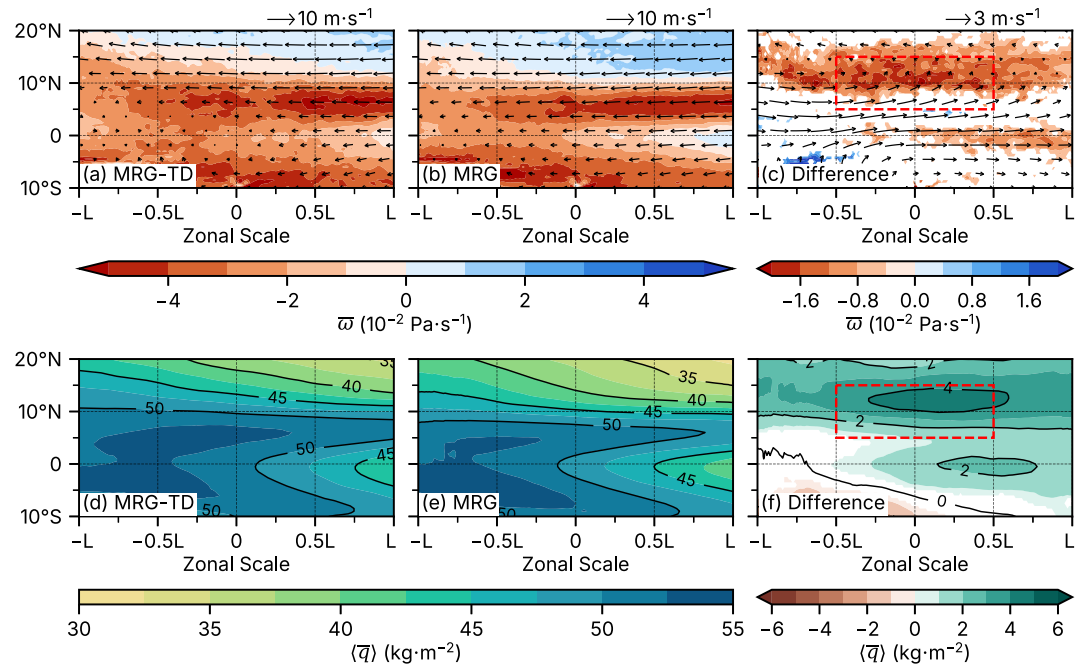


Figure 9. Background horizontal wind at 850 hPa (a–c, vector, $\text{m}\cdot\text{s}^{-1}$), vertical velocity at 850 hPa (a–c, shaded, $10^{-2} \text{ Pa}\cdot\text{s}^{-1}$) and integrated specific humidity ($\bar{q}$) (d–f, contour and shaded, $\text{kg}\cdot\text{m}^{-2}$) of (a, d) MRG-TD type, (b, e) mixed Rossby-gravity (MRG) type and (c, f) their differences when MRG wave is located at 160°E . Only background vertical velocity and integrated specific humidity that are statistically significant at the 90% confidence level are shaded in differences. The red dashed boxes in (c and f) indicate the averaging zone of profile in Figure 10.

horizontal axis represents the longitude of the MRG wave center, as the MRG wave propagates from the rightmost side (170°E) to the leftmost side (155°E). Note that the precipitation term is inverted for comparison with the convective terms. Convective activity dominates the net vertical moistening in both types, as represented by the anomalous vertical moisture transport term $-\omega'\partial\bar{q}/\partial p$ and the anomalous precipitation term $-P'$. The variation of these two terms is highly consistent, indicating that most of the moisture in precipitation originates from anomalous vertical moisture transport.

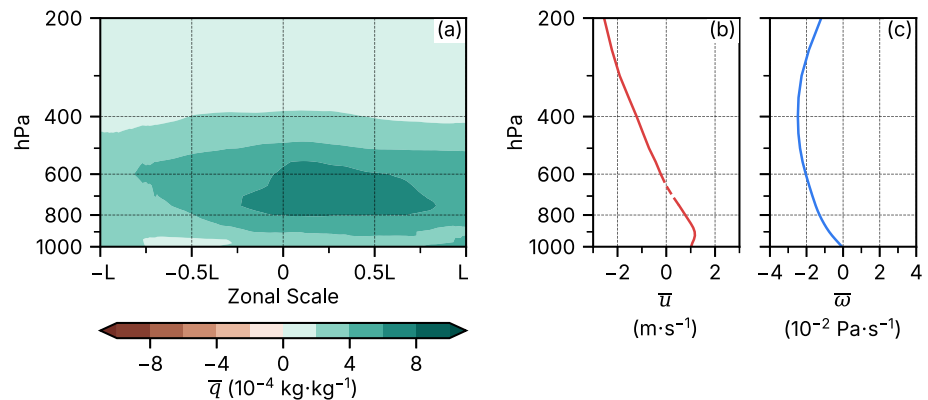


Figure 10. Difference (MRG-TD type minus mixed Rossby-gravity (MRG) type) of (a) background specific humidity $\bar{q}$ (shaded, averaged in 5°N to 15°N , $10^{-4} \text{ kg}\cdot\text{kg}^{-1}$), profile of (b) background zonal wind $\bar{u}$ ($\text{m}\cdot\text{s}^{-1}$) and (c) background vertical velocity $\bar{\omega}$ ($10^{-2} \text{ Pa}\cdot\text{s}^{-1}$) averaged in $-0.5L$ – $0.5L$, 5° – 15°N (red dashed boxes in Figures 9c and 9f) when MRG wave is located at 160°E . Only background specific humidity, zonal wind and vertical velocity that are statistically significant at the 90% confidence level are shaded or plotted with solid lines.

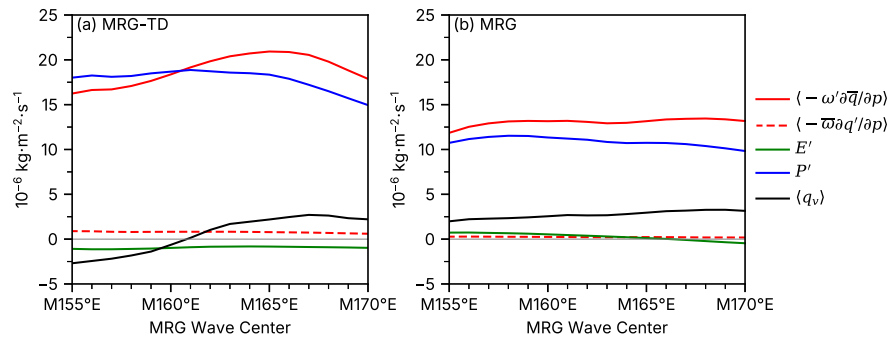


Figure 11. The variation of each integrated term ($10^{-6} \text{ kg} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$) of Equation 8 in north of the mixed Rossby-gravity (MRG) wave center (5° – 15°N , normalized zonal scale $-0.5L$ – $0.5L$, red dashed boxes in Figure 7) during the propagation of (a) MRG-TD type and (b) MRG type. Precipitation term $-P'$ is inverted for comparison with other terms. The M longitude markers indicate the center of MRG wave.

In the MRG-TD type (Figure 11a), anomalous vertical moisture transport first intensifies as the MRG wave propagates from 170°E , reaches a peak around 165°E , and then decreases. The anomalous precipitation term increases gradually, exhibits relatively small variations after 164°E , and exceeds the anomalous vertical moisture transport near 161°E . This crossing of the two terms indicates that the net vertical moistening switches from positive to negative near 161°E , indicating that column-integrated atmospheric moisture accumulates prior to 161°E . As precipitation intensifies and anomalous vertical moisture transport weakens, atmospheric moisture begins to decrease after 161°E due to consumption by condensation. This intensified anomalous precipitation with associated moisture consumption drives the net vertical moistening to turn negative, and this process is a system-wide characteristic of MRG-TD type disturbances. The reduction in anomalous vertical moisture transport after 165°E is attributed to the northwestward shift of the anomalous upward motion center, which moves away from the statistical region north of the MRG wave center.

In contrast, anomalous vertical moisture transport and anomalous precipitation in MRG type (Figure 11b) exhibit relatively small variations throughout the propagation process, and the magnitudes of both terms are smaller than those in the MRG-TD type. The net vertical moistening in MRG type disturbances is also consistently low-amplitude and positive, reflecting a sustained MRG wave propagation process with no transition to TD-type waves. The remaining terms, namely the background vertical moisture transport term $-\bar{\omega}\partial q'/\partial p$ and the anomalous evaporation term E' , make negligible contributions to the vertical moisture budget for both types.

Figure 12 illustrates the vertical distribution of anomalous vertical moisture transport term for the two types of MRG waves, as well as their differences. As discussed earlier, the MRG-TD type disturbance exhibits stronger and deeper upward motion compared to the MRG type. Enhanced background moisture is observed in the lower troposphere between 5°N and 20°N (Figure 9f). The anomalous upward motion in the MRG-TD type disturbances exhibits greater intensity. The off-equatorial moisture anomaly in the lower levels amplifies the mid-level vertical moisture gradient, facilitating enhanced moisture transport to mid-upper levels via convective activity. These two factors contribute to the stronger and deeper anomalous vertical moisture transport in the MRG-TD type disturbance, with the convection center gradually displaced northward compared to the MRG type (Figures 12i–12l). In contrast, the position of anomalous convection in the MRG type remains close to the equator throughout the propagation process, with the majority of the anomalous convection located south of 10°N (Figures 12e–12h). This intense anomalous convection at higher latitudes in the MRG-TD type disturbance is likely a key factor contributing to the northward deviation of the MRG wave during the transition.

4.3. Stronger Moisture Northward Advection in the MRG-TD Type

All terms in the moisture budget Equation 9 are presented in Figure 13, including the net vertical moistening term and the horizontal advection components. Both types of MRG wave disturbances exhibit intensifying anomalous moisture to the north of the MRG wave center during propagation. The moistening rate is similar for the MRG-TD type and the MRG type. The anomalous moisture tendency exhibits a decreasing trend across the propagation of both MRG and MRG-TD type waves. In both types, the anomalous moisture tendency north of the MRG wave center is dominated by three terms: the net vertical moistening q_v , the background zonal advection $-\bar{u}\partial q'/\partial x$, and

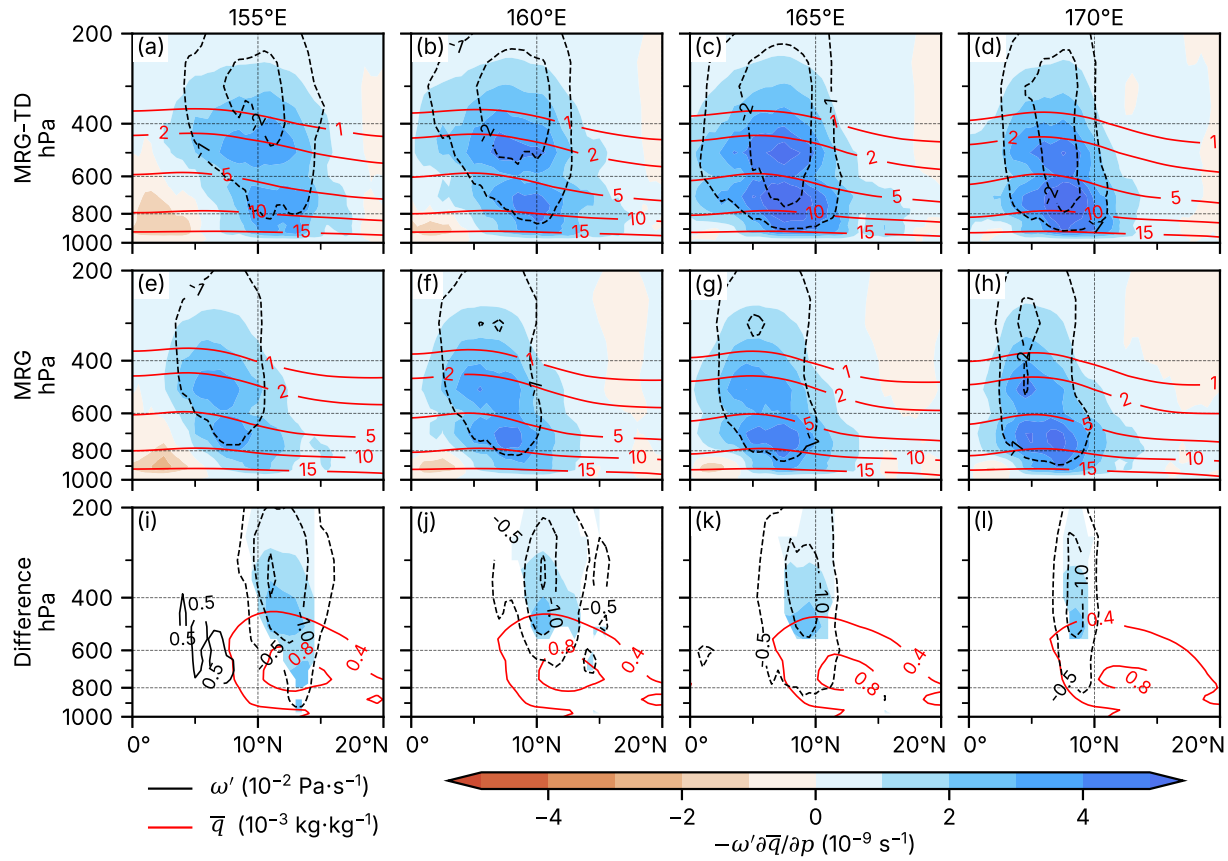


Figure 12. Averaged vertical distribution in $-0.5L$ to $0.5L$ normalized zonal scale of anomalous vertical moisture transport term $-\omega' \partial \bar{q} / \partial p$ (shaded, 10^{-9} s^{-1}), anomalous vertical velocity ω' (black contour, $10^{-2} \text{ Pa} \cdot \text{s}^{-1}$) and background specific humidity $\bar{q}$ (red contour, $10^{-3} \text{ kg} \cdot \text{kg}^{-1}$) of (a–d) MRG-TD type, (e–h) mixed Rossby-gravity (MRG) type and (i–l) their differences when MRG wave is located at (a, e, i) 155°E , (b, f, j) 160°E , (c, g, k) 165°E and (d, h, l) 170°E . Only anomalous vertical moisture transport terms that are statistically significant at the 90% confidence level are shaded in differences.

the anomalous meridional advection $-v' \partial \bar{q} / \partial y$. The net vertical moistening shows a distinct difference between the two types: for the MRG-TD type, it transitions from positive to negative near 161°E , as mentioned previously. The magnitudes and variations of the background zonal advection and anomalous meridional advection are similar between the two types, while the latter exhibits a more pronounced increasing trend for the MRG-TD type and is stronger than its counterpart in the MRG type after the MRG wave propagates past 161°E . In addition, the anomalous zonal advection term $-u' \partial \bar{q} / \partial x$ and the background meridional advection term $-\bar{v} \partial q' / \partial y$ are relatively small for both types, and can be neglected.

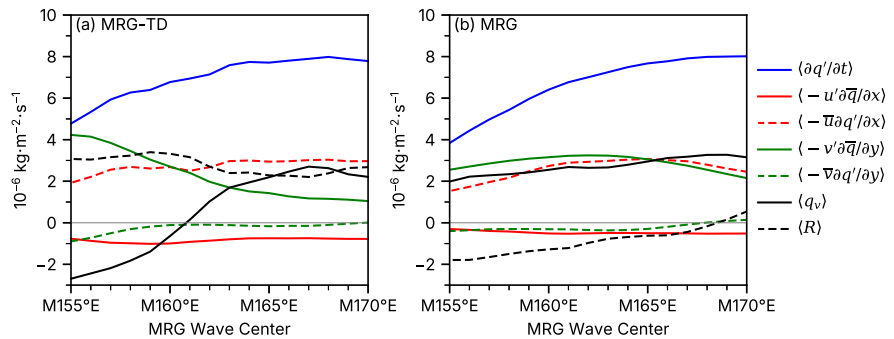


Figure 13. The variation of each integrated term ($10^{-6} \text{ kg} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$) of Equation 9 north of the mixed Rossby-gravity (MRG) wave center (5° – 15°N , normalized zonal scale $-0.5L$ – $0.5L$, red dashed boxes in Figure 7) during the propagation of (a) MRG-TD type and (b) MRG type. The M longitude markers indicate the center of the MRG wave.

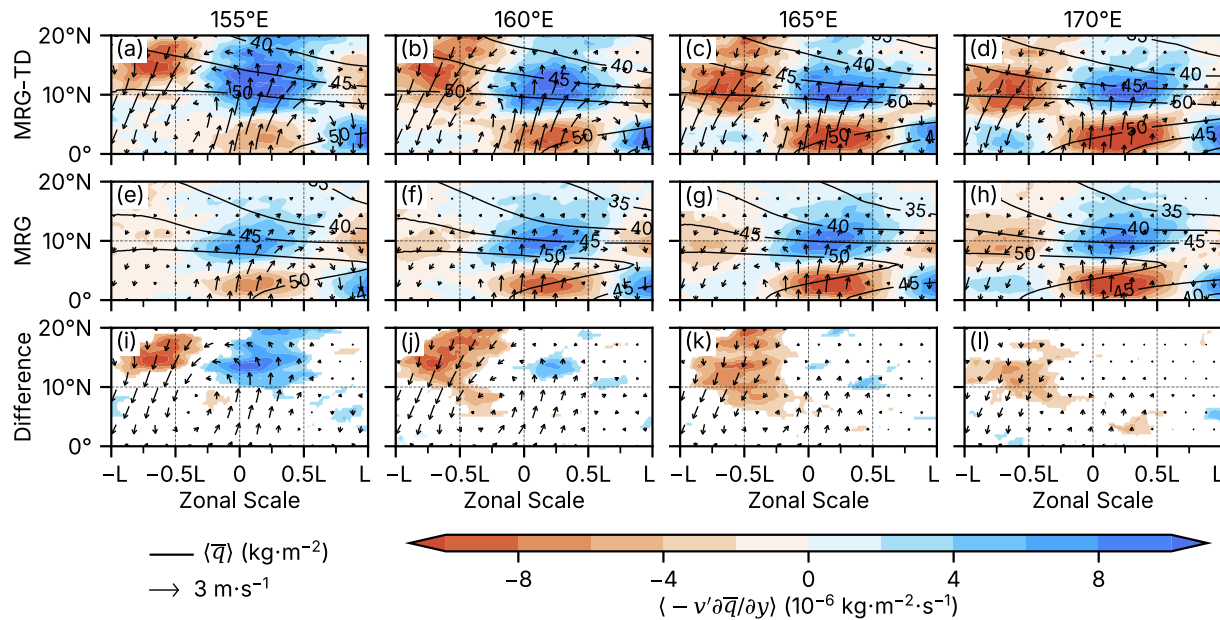


Figure 14. The spatial distribution of integrated anomalous meridional advection term ($\langle -v' \partial \bar{q} / \partial y \rangle$) (shaded, $10^{-6} \text{ kg} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$), integrated background specific humidity ($\bar{q}$) (contour, $\text{kg} \cdot \text{m}^{-2}$) and anomalous wind at 850 hPa (vector, $\text{m} \cdot \text{s}^{-1}$) of (a–d) MRG-TD type, (e–h) mixed Rossby-gravity (MRG) type and (i–l) their differences when MRG wave is located at (a, e, i) 155°E, (b, f, j) 160°E, (c, g, k) 165°E and (d, h, l) 170°E. Only integrated anomalous meridional advection terms that are statistically significant at the 90% confidence level are shaded in differences.

Figure 14 illustrates the spatial distribution of the anomalous meridional advection term, background moisture, and anomalous wind at 850 hPa for both types of MRG waves. Distinct differences exist in the background moisture distribution between the two MRG wave types. The MRG-TD type exhibits stronger background moisture north of 5°N than the MRG type (Figures 9d–9f). The background moisture field features a zonally oriented ridge near 6°N, leading to a negative meridional moisture gradient north of the ridge and a positive gradient to its south. For the MRG-TD type, enhanced background moisture north of 5°N, leading to the meridional moisture gradient that is relatively weaker than that for the MRG type in this region. Meanwhile, the anomalous meridional wind for the MRG-TD type is significantly stronger than that for the MRG type, driving stronger northward moisture advection from the moisture ridge east of the MRG center despite the weaker meridional background moisture gradient (Figures 14i–14l). The stronger southerly wind anomaly associated with the MRG-TD type not only advects more background moisture northward, which amplifies the moisture tendency north of the MRG wave center and contributes to the northward shift of the moisture anomaly, but also induces ascending motion and promotes the development of more intense convection for this wave type.

Note that for the MRG-TD type, the anomalous northerly wind within the cyclonic circulation west of the MRG wave drives intense negative moisture advection north of the moisture ridge (Figures 14i–14l). This spatial pattern of anomalous meridional advection, characterized by negative values ahead of the MRG wave and positive values behind the wave, reduces the westward propagation speed of the anomalous moisture perturbation associated with the MRG wave, which may also contribute to the phase speed deceleration of the MRG wave and the MRG-TD transformation.

Another important term in the moisture budget is the background zonal advection term $-\bar{u} \partial q' / \partial x$. For both MRG wave types, this term exhibits an overall decreasing trend as the MRG wave propagates westward, while the term for the MRG type first increases and peaks near 165°E.

Figure 15 illustrates the spatial distribution of the background zonal advection term, anomalous moisture, and background wind at 850 hPa for both types of MRG waves. The difference in background zonal wind between the two types is that the background easterly wind for the MRG-TD type is relatively weaker south of 10°N, which advects less moisture in this near-equatorial region, favoring the northward shift of the anomalous moisture center of the MRG wave. Meanwhile, the spatial distribution of anomalous moisture is also similar between the two types, exhibiting a southwest-northeast tilted pattern. However, the higher anomalous moisture content for the

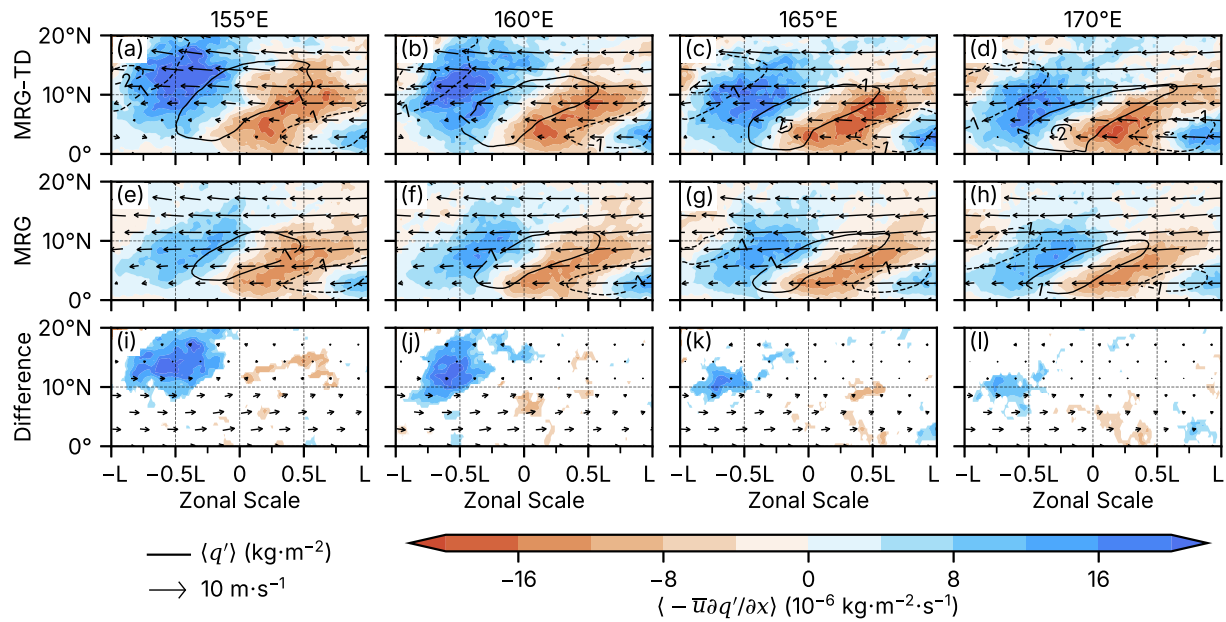


Figure 15. The spatial distribution of integrated background zonal advection term ($-\bar{u}\partial q'/\partial x$) (shaded, $10^{-6} \text{ kg} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$), integrated anomalous specific humidity (q') (contour, $10^{-1} \text{ kg} \cdot \text{m}^{-2}$) and background wind at 850 hPa (vector, $\text{m} \cdot \text{s}^{-1}$) of (a–d) MRG-TD type, (e–h) mixed Rossby-gravity (MRG) type and (i–l) their differences when MRG wave is located at (a, e, i) 155°E , (b, f, j) 160°E , (c, g, k) 165°E and (d, h, l) 170°E . Only integrated background zonal advection terms that are statistically significant at the 90% confidence level are shaded in differences.

MRG-TD type and its northward shift during MRG wave propagation lead to a stronger background zonal moisture advection for the MRG-TD type. This anomaly moves northward along with the anomalous moisture displacement in the MRG-TD type as the MRG wave propagates from 170° to 155°E (Figures 15i–15l). In addition, as MRG waves propagate from the central to western Pacific, the background easterly wind weakens gradually, which is reflected in the decreasing trend of the background zonal advection term for both types.

5. Conclusion and Discussion

This study investigates the moisture variation during the transition of MRG wave to TD-type wave over the western North Pacific. An improved identification method for MRG wave events is developed to detect individual MRG wave events during the 1979–2025 study period. Valid individual MRG wave events are further classified into two categories: the transformed MRG-TD type and the non-transformed MRG type. For the transformed MRG-TD type, the MRG wave disturbances deviate from the equator, and evolve into TD-type disturbances; the non-transformed MRG disturbances are trapped in the equatorial region. Non-transformed MRG waves are most active from March to June and November to December, while transformed MRG waves are less frequent, with their occurrence peaking from July to December. The interannual variation of non-transformed MRG waves exhibits a weak increasing trend, which is consistent with previous studies (Raghavendra et al., 2019), while a significant decreasing linear trend is detected for transformed MRG waves.

To identify the factors influencing the MRG wave transition, this study compared the structure and moisture budgets between the transformed and non-transformed MRG types. For the transformed type, background conditions exhibit enhanced moisture and westerly winds north of 5°N in the lower troposphere, accompanied by intensified background vertical motion from the lower to the upper troposphere, relative to the non-transformed type. As the MRG wave propagates west of 170°E , the transformed MRG-TD wave disturbance exhibits a stronger anomalous southerly wind in the lower troposphere, resulting in enhanced anomalous convergence and upward motion in the region of the northward outflow. Moreover, this intensified anomalous southerly wind advects more moisture northward from the ridge of background moisture near 6°N , humidifying the off-equatorial region and promoting convection development. In the non-transformed MRG wave, the weaker anomalous southerly wind results in reduced moisture north of the MRG wave center. Furthermore, the vertical tilt of the transformed MRG-TD wave decreases as it propagates westward. In contrast, the vertical tilt of non-transformed

MRG waves decreases more slowly, accompanied by weaker southerly winds and upward motion in the troposphere.

In summary, for the transformed MRG-TD type waves, a convection-favorable background environment, anomalous northward moisture advection, and upward motion anomaly collectively contribute to the humidification and development of stronger convection in off-equatorial regions, driving the northward deviation of the coupled circulation and MRG disturbance. This result indicates that the wind field structure and moisture characteristics of MRG wave disturbances are important factors governing the evolution and transition of MRG waves, in addition to the background dynamic environmental conditions.

Previous studies have suggested that the transition of MRG waves is influenced by the monsoon trough (Chen, 2012; Chen & Huang, 2009; B. Wang & Xie, 1996; Wu et al., 2014, 2015), which modifies the background conditions and affects the transition process. However, the monsoon trough varies seasonally, while the MRG wave transition events observed occur throughout the year. Our current study suggests additional features and associated processes for the MRG-TD transformation. The transformed MRG-TD type waves are significantly influenced by a convection-favorable background environment, anomalous northward moisture advection, and upward motion anomalies. Together, these factors enhance humidification and foster stronger convection in off-equatorial regions, leading to a northward deviation of the MRG wave disturbances. The interaction between MRG waves and the background circulation, a potential factor influencing MRG wave transition, is not considered in the present study, and warrants further investigation through numerical experiments in future research.

The analysis conducted in this study is constrained to the ERA5, MSWEP, and GPM data sets; consequently, a degree of uncertainty remains in the conclusions drawn. Previous studies have demonstrated consistency in the representation of tropical waves across different reanalysis data sets (Fujiwara et al., 2012; Kim et al., 2019). Nonetheless, discrepancies remain within the reanalysis data sets due to parameterization schemes. Additionally, the residual term R in the moisture budget for the MRG-TD type is non-negligible, likely due to subgrid-scale and nonlinear processes associated with stronger convection activity. Thus, convective-permitting simulations are required to adequately resolve these processes.

Appendix A: MRG Wave Tracking and Composites

The method used to identify, classify, and composite MRG waves in this study is summarized in Figure A1 and as follows:

1. Apply a 3–8-day band-pass filter and an 8-day low-pass filter to obtain anomalous and background variables, respectively.
2. Identify MRG wave events over the western Pacific using the spatiotemporally filtered 850 hPa meridional wind, following the method described in Section 2.2.
3. MRG wave events that propagate from east of 170°E to west of 150°E are first selected. When these waves cross 160°E, the corresponding 3–8-day band-pass-filtered 850 hPa meridional wind fields (covering 100°E–180°, 20°S–20°N) at timestamps lagged by 54 hr are generated via linear interpolation and extracted. Each MRG wave event and its associated wind field constitute one sample for subsequent clustering analysis.
4. Apply K-means clustering to group the samples into two clusters based on the spatial pattern of the meridional wind. The clustering output is represented by type-labeled MRG wave events.

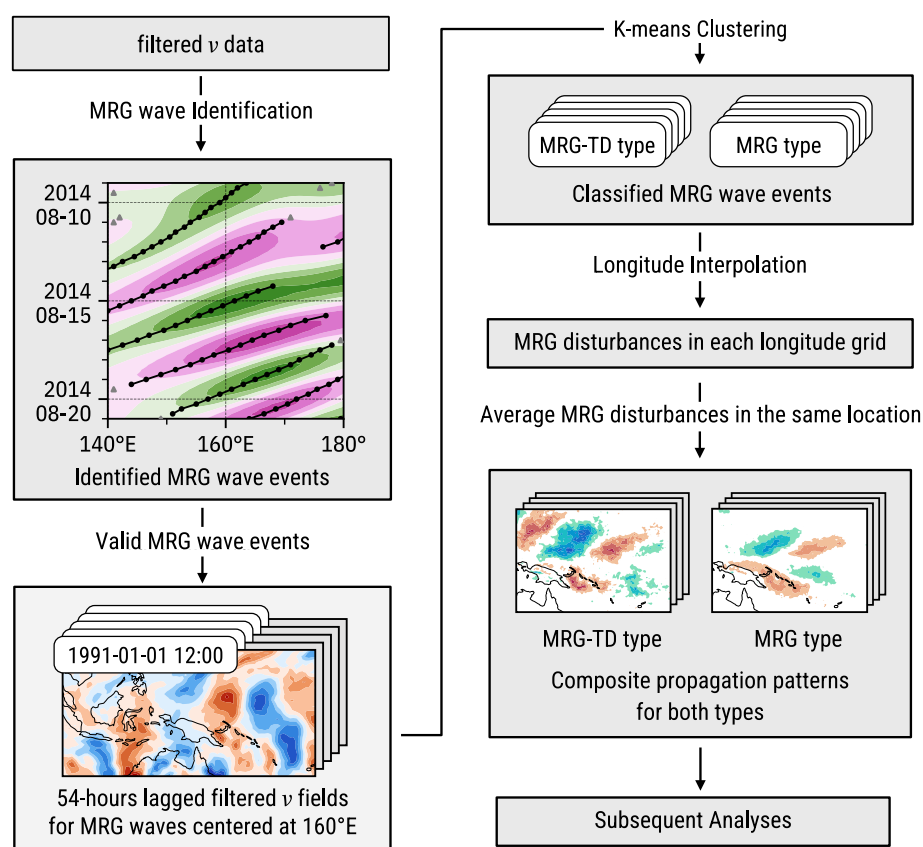


Figure A1. Flowchart of the data processing in this study.

5. For each cluster type, compute composite anomalies by averaging anomalous variables at the lead-lag timestamps when the MRG wave is located at 160°E (Figure 2), to obtain the evolution of the two MRG wave types.
6. Variables are linearly interpolated at 1° longitude intervals as each MRG wave event propagates from 170°E to 155°E. Because the wavelength of each individual MRG wave differs, the corresponding variables are scaled in the zonal direction to a normalized zonal coordinate system based on the wavelength of the respective MRG wave.
7. Variables at the same longitude points are averaged for each MRG type. The obtained averaged results characterize the propagation process of the two MRG wave types at the specified location.

Table B1

The Number of Associated TC Genesis When Mixed Rossby-Gravity Waves Are at Different Longitudes

Type	155°E	160°E	165°E	170°E
MRG-TD (360)	15.3% (44)	21.2% (61)	14.6% (42)	15.6% (45)
MRG (927)	8.4% (74)	8.0% (70)	8.3% (73)	9.2% (81)

Note. Percentage values indicate the ratio of TC genesis events to the total number of MRG waves of each type, with the corresponding TC genesis count given in parentheses. The total number of MRG wave events for each type is shown in parentheses in the first column. Abbreviation: MRG, Mixed Rossby-gravity.

Appendix B: TC Genesis Statistics

TC genesis is defined as the first track point in the IBTrACS data set, and only the genesis in the northwestern Pacific (100°E–180°, Northern Hemisphere) during 1979–2025 is considered.

TC genesis events occurring within 24 hr after an MRG wave propagates to the specified longitudes (155°E, 160°E, 165°E, and 170°E) are selected. The longitudes of these selected events are converted to a normalized zonal scale based on the wavelength of the associated MRG wave. These events are plotted as green dots in Figure 5.

The corresponding number of TC genesis events (the number of green dots in Figure 5) at each MRG wave center longitude are summarized in Table B1.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The ERA5 reanalysis data used in this study are obtained from the European Centre for Medium-Range Weather Forecasts and are available at <https://cds.climate.copernicus.eu/>. The GPM precipitation data used in this study are obtained from NASA GPM mission and are available at <https://gpm.nasa.gov/data>. The MSWEP precipitation data used in this study are obtained from GLoH2O and are available at <https://www.gloh2o.org/>. The Nino 3.4 index data used in this study are obtained from the NOAA physical sciences laboratory and are available at https://psl.noaa.gov/data/timeseries/month/Nino34_CPC/. The IBTrACS data set used in this study is obtained from NOAA National Centers for Environmental Information and is available at <https://www.ncei.noaa.gov/products/international-best-track-archive>.

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